

COMP3322 Modern Technologies on World Wide Web

Group Project – Develop Your Web Application (30%)

1. General Overview

In this group project, teams of 4-5 students will collaboratively design, develop, test, deploy and present a web application of your own proposed idea.

2. Minimum Project Requirements

The team's self-proposed project should achieve the following minimum requirements:

1. Frontend - Single Page Application (SPA)

- Choose one of the following frameworks:
 - Angular
 - React
 - Vue3
- Not allowed: Expo, React Native, NativeScript, or any other cross-platform mobile framework.
- Your frontend implementation must be responsive.

2. Backend - RESTful API

- Use Node.js + Express as backend framework.
- The backend must expose a RESTful API.
- Use correct HTTP status codes appropriately (200, 201, 400, 401, 403, 404, 500, etc.).

3. Database - MySQL

- Can only use MySQL.

4. Deployment Requirements (TBD)

- Deploy on a Linux virtual machine using Docker Compose.
- All deployment-related files must be committed to GitHub, such as:
 - Dockerfile
 - docker-compose.yml
 - README.md with detailed deployment instructions
 - And examiner should be able to deploy it without error
- Security: Do not commit any real secrets or credentials (e.g., .env with production keys, database passwords, API secrets). Use environment variables or example template files.
- Details will be shared with you later, we are still working with ITS in setting up environment.

5. Complexity Expectations

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- The project must demonstrate sufficient system complexity, as judged by the instructor.
- To help you gauge the required depth, consider (these are example only, not mandatory checkboxes):
 - Database Design Complexity
 - Real-time or Asynchronous Processing
 - Data Visualization
 - Aesthetics
 - Meaningful business logic beyond simple CRUD
 - OAuth2.0 (bonus)
 - etc.

The project should meet all the above requirements to achieve a passing grade for the group project component of the course.

3. Team Creation

Spend some time talking to your classmates in the first two weeks and form a group. Then record your group members' full names, student numbers, and email addresses.

Submit all your group members' information (full names, email addresses, and student numbers) via Moodle before 7:00 PM, 15 September 2026. Only one submission is required per team. Path: COMP3322 Course Page → Project → Group Formation

If you do not belong to any group, we will assign you to a random group. We will publish all grouping information as an Excel file on Moodle on the submission deadline day. Please review the grouping information carefully to check for any typos. And let us know during tutorial time.

4. Proposal (5%)

Create a proposal folder at the root of your project repository. Place your proposal inside a Proposal.md file within this folder. Your proposal will be reviewed during the Alpha Version Interview.

Your proposal must include all original baseline items below (2%):

- Member Information: Full name and student number for each team member
- Project title
- Project Description: One paragraph summarizing your project and identifying your target end-users
- Feature list, strictly split into two categories:
 - Must-have: Non-negotiable core features your project must complete

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- Nice-to-have: Optional bonus features; only implement these after all must-have items are finished
- Full technology-stack selection: frontend framework, backend framework, database, deployment platform, and any external third-party things you plan to integrate.
- Preliminary team task allocation: Planned responsibilities for each group member.

Additional required sections to support us better understanding your project (3%):

1. Problem & Target User Context
 - Briefly explain what real-world small problem your web application intends to solve. Describe who your users are and why a web-based application is a good fit for this use case. You may briefly mention simple similar websites as reference; deep competitive analysis is not required.
2. High-level Workflow Description
 - Describe in plain text how end-users interact with your web application. Walk through typical user journeys: what actions users take, what pages they visit, and what responses the application provides to them. You **do NOT need** to describe backend-to-database data flow details.
 - You may attach a simple sketch (hand-drawn photo; simple PPT drawing screenshot is acceptable). Your sketch should illustrate major user-facing pages/components and user transitions between them. Professional-quality system diagrams are not expected.
3. Anticipated Learning Challenges & Self-assessment
 - List 2-3 realistic challenges your group expects to face during development. Since most students only have Python experience and are new to frontend, Node.js/Express, MySQL and Docker deployment, your challenges should reflect your actual learning curve. For each challenge, write one simple practical plan on how your group will try to overcome it.

5. Alpha & Beta Version

You will complete two intermediate versions prior to your final version. You must book a slot during tutorial time (Friday 12:00 – 12:50) for your interview. Each interview will last approximately 8-10 minutes and is intended to verify your understanding of the features you have implemented. We will ask every student questions.

A maximum 5% deduction per version will be applied to your project mark (applied to individual marks, not group marks) in the following cases:

- Absence from interview (e.g., failure to book an interview slot, no-show for your booked timeslot)
- Failure to provide a satisfactory explanation of the feature and code during the interview
- No submission on Moodle (this applies to the whole group)

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	Alpha	Beta
Minimum Deliverables Requirements	1. Complete the project proposal 2. Complete database schema implemented and wired into backend 3. Backend API skeleton built; subset of core features working end-to-end (frontend ↔ backend ↔ database) 4. Corresponding frontend UI pages for implemented features 5. All Alpha-version source code committed and pushed to team GitHub repository (using tagging)	1. All core features defined in the proposal fully implemented 2. Basic client-side & server-side input validation; major critical bugs fixed 3. All beta-version source code committed and pushed to team GitHub repository (using tagging) 4. Application deployed (this would be updated after we settle down things with ITS)
Deliverables Submission Deadline	Submit your source code in a zip file to Moodle submission box before your interview start time	Submit your source code in a zip file to Moodle submission box before your interview start time
Interview Booking	The activity will be opened on Moodle on 28 September	The activity will open on Moodle on 26 October
In-class Interview	2 Oct and 9 Oct, 12:00-12:50 (tutorial session time)	30 Oct and 6 Nov, 12:00-12:50 (tutorial session time)

6. Final Version (20%) & In-class Presentation (5%)

Final Version Requirements:

1. All core features fully polished; optional features completed per agreed complexity.
2. All major and minor bugs fixed; application runs stably in the production environment.
3. Final documentation (PDF stored in the root of repository), for example:
 - a. Project title
 - b. Project description
 - c. Feature list
 - d. Team task allocation
 - e. System architecture diagram
 - f. Database ER diagram
 - g. API Document
 - h. Other relevant supporting materials
4. Final stable deployment maintained and publicly accessible; ensure the URL is publicly reachable.
5. Complete the most up-to-date project README.md (must include the public deployment URL).

Presentation Rules

In-class presentations will take place on 20 November (12:00-12:50), 23 November (12:00-13:50), and 27 November (12:00-12:50; this slot is subject to the final number of groups).

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Presentations must be delivered fully off script. You are not allowed to read directly from any form of script, printed cue cards, notes, mobile phones, or other electronic devices during their presentation.

Reading from scripts, cue cards, or mobile devices during your presentation will result in a mark deduction of up to 5% of the total project grade. This penalty applies to individual presenters who read, even if other group members present without a script.

You may use your presentation slides as visual prompts only. Slides should contain key points and visuals, not full sentence speaker scripts. Do not paste large blocks of text onto slides to serve as a reading script.

7. Grading

The project accounts for up to 30% of your overall course grade, broken down as follows:

- Project Proposal: 5%
- Presentation: 5%
- Final Version: 20%

The Final Version will be assessed against the following dimensions:

- Fulfilment of minimum project requirements
- Application stability and functional performance
- Adherence to best practices and design principles
- UI/UX quality
- Final Documentation Quality

Important Note:

- We will only test and grade your application based on the production environment (please wait for further instruction on deployment). If your application does not work on production, your project will likely receive a low score.
- The final deployed code should be the exact same as the submitted source code on GitHub.
- The project should meet all minimum project requirements to achieve a passing grade for group project component of the course.
- All team members must put their best effort into contributing to the project (git contribution: commit, pull request). We will justify your personal project grade based on your individual contributions.

8. Academic Integrity

The course academic integrity policy applies to this project. All code for the project must be written only by your team members, with proper use of AI tools.

If you use any UI components or code snippets from online sources, you must clearly cite them on a credits page inside your application.

You may choose your own project topic, but these rules must be followed:

- Do not reuse code you wrote outside this course / written for any work / written for another course etc.
- You cannot receive payment for working on this same project for another organization while this project is ongoing.
- Every team member is responsible for the work they commit to the project repository.